



OCB 16.07.2026

Bourns DMS 5/5.2_Dual source for "Torque Terminal"
and Reduce Pd/Au plating → **Subsupplier changes**

ECR Number: **3E10183210**

Date: 2026-07-16 / 2026-07-21

Past request → new situation (Re-cap)

- In **the past** the team introduced the topic as second supplier for the torque sensor.

Change Description:

Type of change: **Introduction of second sub-supplier** (dual sourcing) and Material weight change (selective plating)

Current: Full Pd/Au plating on torque terminal (single source)

New: Reduced Pd/Au selective (WBO) plating; dual sourcing introduced

Tier Level: Tier 3 (sub-supplier at Bourns)

Motivation / Benefit:

Cost reduction by reducing precious metal usage
(in case of implementation w/o customer involvement
the estimated yearly savings would be nearly 50k EUR)

Improved supply chain robustness via dual sourcing

Affected product / component / plants:

VM Products: EPS APA, DP

Components: DMS5 TIS/TOS, DMS5.2

Bosch Plants: MrP (HU), SgmP (DE), FleP (US)

Field Experience: The torque terminal source is used in the DMS6 AD program (wirebond only plating)

- **New situation:** change in sub-supplier chain.

Metalis USA was acquired by Leman Industrie, and it has been decided to **relocate its manufacturing** operations from Denville, New Jersey, USA to Rochester Hills, Michigan, USA

Transfer in two steps: 1st in July

2nd from September to November

Motivation: Rationalize fixed cost and build North America headquarters

Based on requirements the new location must be validated (validation can begin after the transfer)

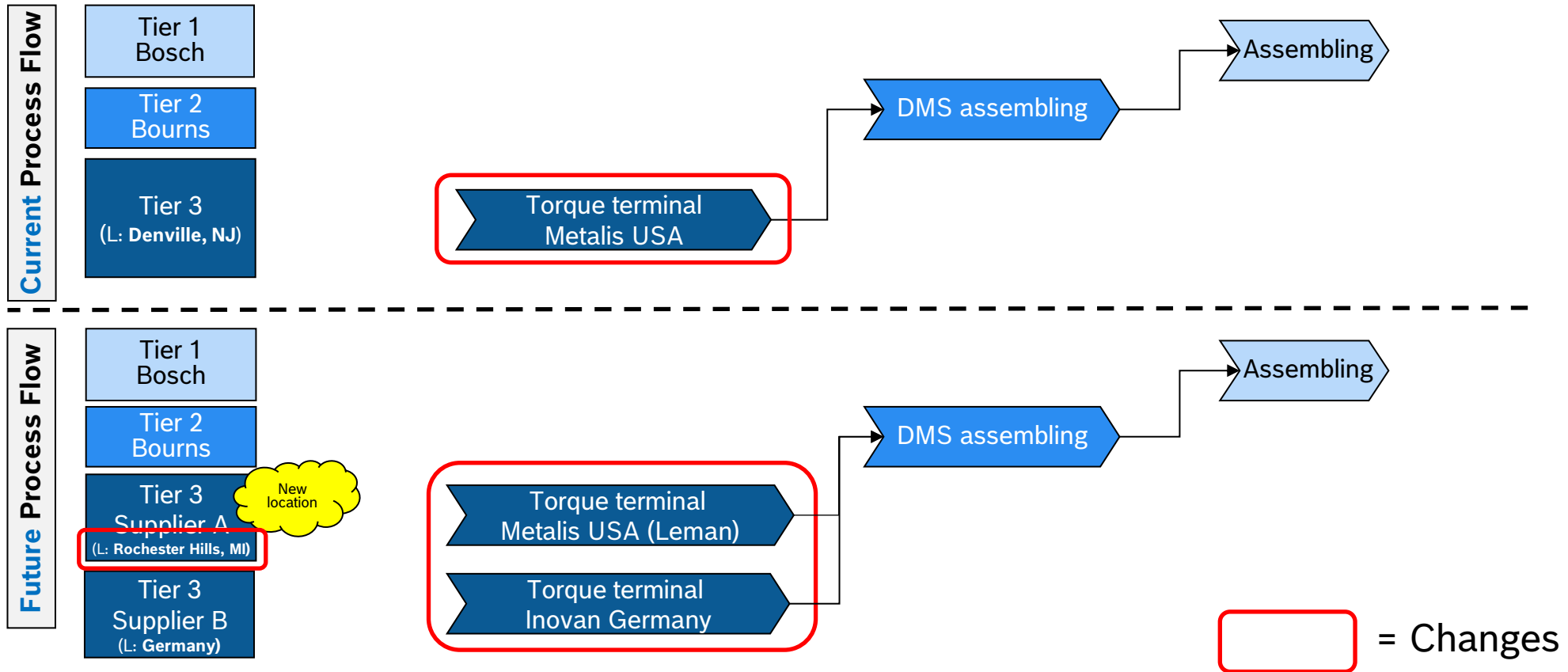
The latest date of approval expected by the supplier: **cw48,2026***



Change is re-discussed on Mat3rd and CCB.

Team proposed to inform the customer VW about the new situation including the WBO plated terminal as second source.

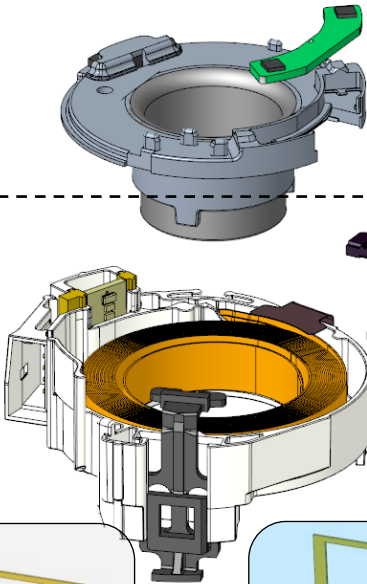
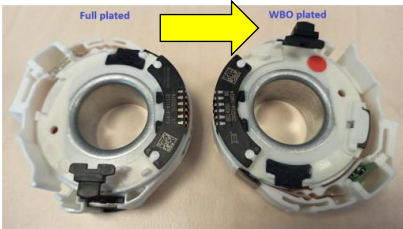
Process / Logistic flow (current vs. future)



What? How? Where?

WBO* (**Wire Bond Only**) means, the terminal is plated only on the functional important area, where the gold wire bonding process requires the presence of gold.

DMS6 PV6 has the same change, VW is informed. PV passed, VW vehicle testing is passed as per BHVP26-65 report. See below:

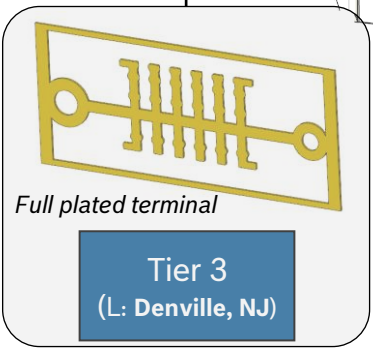
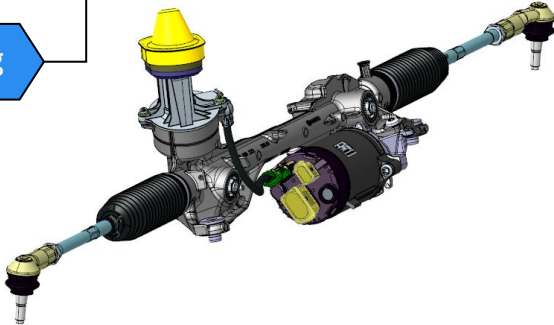


DMS assembling

Tier 2
Bourns,
(L: Hungary, Ajka,
Mexico Tijuana)

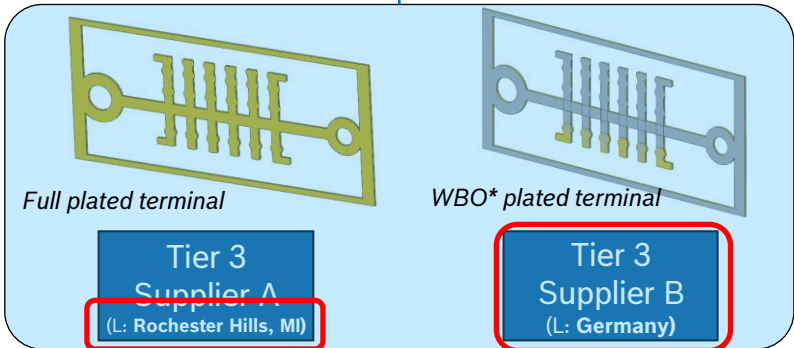
Steering assembly

OEMs



Tier 3
(L: Denville, NJ)

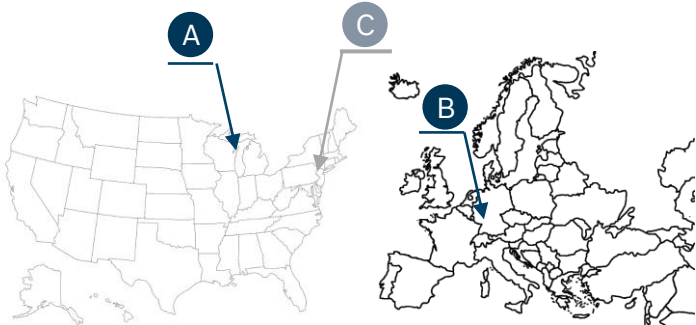
current



Tier 3
Supplier A
(L: Rochester Hills, MI)

Tier 3
Supplier B
(L: Germany)

proposed



VM26_1034_Bourns DMS 5/5.2_Dual source "Torque Terminal" Reduce Pd/Au plating

Decision/Backlog

- Mat 3rd Team requests official decision via ECR-SC – Resp.: Zorell

Mat3rd + CCB:

Since the Review Team has decided that only VW needs to be informed involved ("Information only"), I would strongly prefer implementing the change. Otherwise, there is a risk that Bourns could raise a claim for the investments already made in preparation for this project. Since Bourns did not win the DMS5 auction, the likelihood of such a claim may be even higher if we decide not to introduce the dual source after they have already invested in the necessary equipment and preparations.

I believe we could consider the following approach and line of argumentation:

The current terminal sub-supplier has been acquired by another company. As a precautionary measure and to **strengthen supply chain resilience**, we are introducing a second qualified sub-supplier as a dual source.

The **second supplier has already been fully validated**, as it was planned to be implemented for the lifetime extension project (2028) in any case (only earlier implementation)

This is considered a **low-risk change**, as the **second supplier's terminal has already been successfully used and proven in other sensor variants**.

The terminal is used by Bosch only for programming during manufacturing. It is **not a customer interface** and has no impact on the functionality, performance, or reliability of the sensor in the field.

Based on the above, the **change should be regarded as a risk mitigation and supply assurance measure**.

VM26_1034_Bourns DMS 5/5.2_Dual source "Torque Terminal" Reduce Pd/Au plating

Current situation

- Previous suggestion from OCB to implement with lifetime extension bundle from 2028
- Supplier Bourns escalated and complaint poor support from Bosch:
 - 1st idea pitched in supplier day “Leading the competition” 18.03.2025
 - Feedback Bosch positive and Bourns considered as formal “GO”
 - Implementation started and engineering work and investment in Tier 3 (Approx. 100k €)
 - Review Team has decided that only VW needs to be informed involved ("Information only")
- **Current situation:**
 - The current terminal sub-supplier has been acquired by another company. As a precautionary measure and to **strengthen supply chain resilience**, we are introducing a second qualified sub-supplier as a dual source.
 - The **second supplier has already been fully validated**, as it was planned to be implemented for the lifetime extension project (2028) in any case (only earlier implementation)
 - This is considered a **low-risk change**, as the **second supplier's** terminal has **already been successfully used and proven in other sensor variants**.
 - The terminal is used by Bosch only for programming during manufacturing. It is **not a customer interface** and has no impact on the functionality, performance, or reliability of the sensor in the field.

Backup

VM26_1034_Bourns DMS 5/5.2_Dual source "Torque Terminal" Reduce Pd/Au plating

Change Description Overview

Change Description:

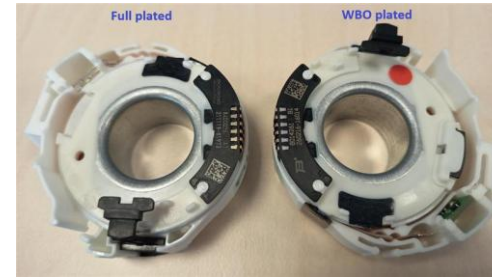
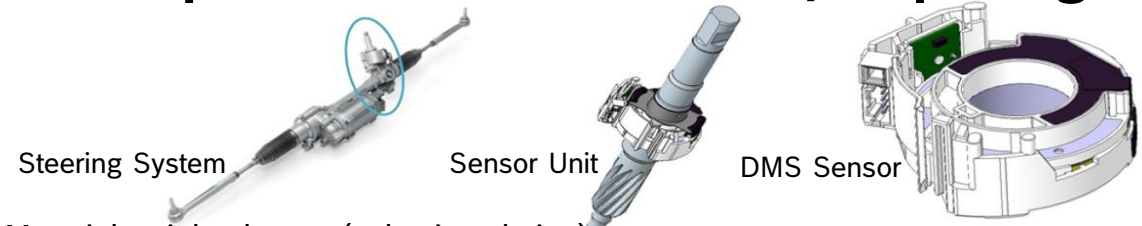
- Type of change: Introduction of second sub-supplier (dual sourcing) and Material weight change (selective plating)
- Current: Full Pd/Au plating on torque terminal (single source)
- New: Reduced Pd/Au selective (WBO) plating; dual sourcing introduced
- Tier Level: Tier 3 (sub-supplier at Bourns)

Motivation / Benefit:

- Cost reduction by reducing precious metal usage (in case of implementation w/o customer involvement the estimated yearly savings would be nearly 50k EUR)
- Improved supply chain robustness via dual sourcing

Affected product / component / plants:

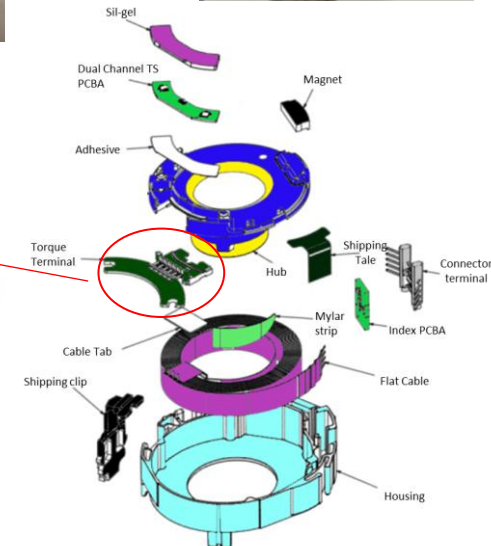
- VM Products: EPS APA, DP
- Components: DMS5 TIS/TOS, DMS5.2
- Bosch Plants: MrP (HU), SgmP (DE), FleP (US)
- Field Experience: The torque terminal source is used in the DMS6 AD program (wirebond only plating)



Actual - full plating: (p/n: 84000-00151)
937,1 mm²

New – WBO plating: (p/n: 84000-13091)
73,6 mm²

Reference:
DMS6 AD program





EVC2600847 Measurement Notes

DMS5 WBO plating thickness

Tóth-Ilkó Ákos
2026.06.24.

EVC2600847 Measurement Notes

Summary

▪ Target of the analysis

Characterize and compare the coating system of the new sample to the old one.

▪ Samples to be analysed

- Old sample, with full plated coating system
- New sample, with partial plated coating system

▪ Applied instruments and methods

- Cyber TECHNOLOGIES CT300 Non-contact profilometer
- M4 Tornado Micro-X-ray Fluorescence Spectroscope (XRF)
- ThermoFisher Scios 2 DualBeam Focused Ion Beam Scanning Electron Microscope (FIB-SEM)

▪ Short results

▪ Layer thickness results

Method	Sample pin	Ni [µm]		Pd [µm]		Au [µm]	
		Average	St dev	Average	St dev	Average	St dev
XRF	New 1	2,99	0,232	0,089	0,0125	0,020	0,0009
	Old 1	1,61	0,034	0,040	0,0033	0,016	0,0019
FIB-SEM	New 1	3,119	0,215	0,0701	0,0033	0,0225	0,0014
	Old 1	1,773	0,081	0,0443	0,0021	0,0223	0,0012
Limits		min 0,76		0,03-0,15		0,005-0,025	

▪ Surface roughness results

Sample pin	Ra (µm)	Rz (µm)
Old	0,54	2,86
New	0,67	4,08

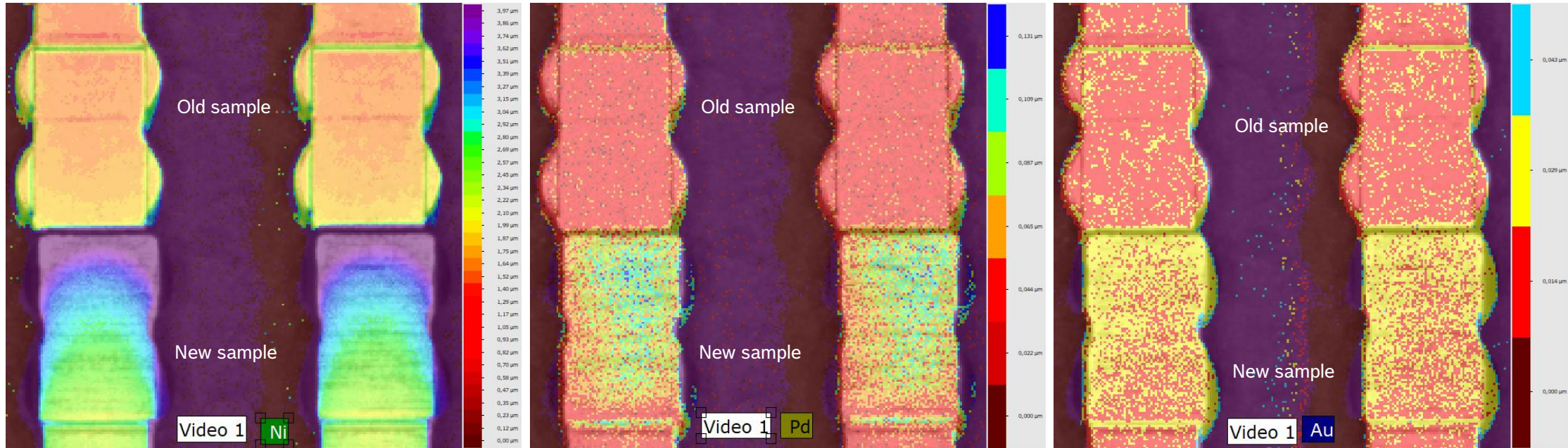
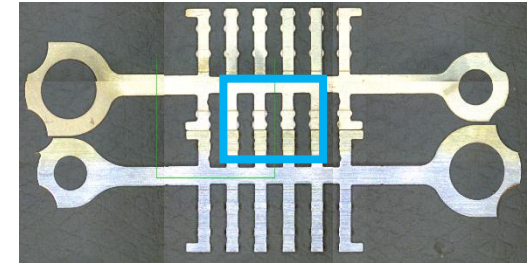
▪ Conclusion

- Most of the cases, a significantly higher layer thickness can be measured on the new sample (within limit at both samples)
- The new sample shows higher values in surface roughness
- The surface of the new sample looks like more homogenous compared to old one

Old sample vs New sample

EVC2600847 Measurement Notes

Sample comparison – XRF coating thickness maps

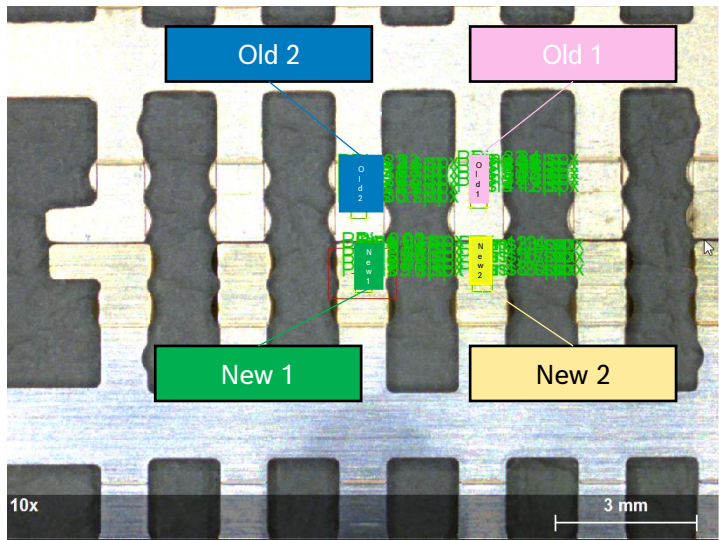


Layer thickness distribution results based on XRF measurements – Nickel, Palladium and Gold coating thickness maps. The colour scale represents the thickness values

- In all three cases (Ni, Pd, Au), a significantly higher layer thickness can be measured on the new sample
- The nickel shows a decreasing layer thickness from the edge to the center on the new sample, then its value stabilizes

EVC2600847 Measurement Notes

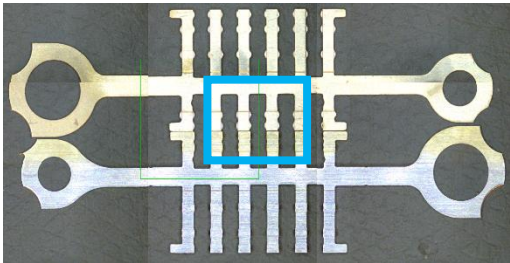
Sample comparison – XRF Multipoint



XRF multipoint measurement locations. The different colours represents the different samples and pins. The top is the „old sample“, the bottom is the „new sample“. As the new sample is rotated, the „old 1“ and „new 1“ represent the same pin position (so „old 2“ and „new 2“)

Spectrum	Substrate		Layer 1		Layer 2		Layer 3		Sample pin
		Cu [%]	Thickn. [µm]	Ni [%]	Thickn. [µm]	Pd [%]	Thickn. [µm]	Au [%]	
Pins 01	Base	100	3,323	100	0,086	100	0,021	100	New 1
Pins 02	Base	100	3,336	100	0,094	100	0,020	100	
Pins 03	Base	100	3,143	100	0,094	100	0,020	100	
Pins 04	Base	100	3,128	100	0,106	100	0,019	100	
Pins 05	Base	100	2,938	100	0,090	100	0,021	100	
Pins 06	Base	100	2,950	100	0,104	100	0,020	100	
Pins 07	Base	100	2,826	100	0,070	100	0,021	100	
Pins 08	Base	100	2,830	100	0,096	100	0,019	100	
Pins 09	Base	100	2,715	100	0,071	100	0,019	100	
Pins 10	Base	100	2,708	100	0,080	100	0,020	100	
Pins 12	Base	100	3,216	100	0,086	100	0,020	100	New 2
Pins 13	Base	100	3,145	100	0,100	100	0,021	100	
Pins 14	Base	100	3,188	100	0,106	100	0,021	100	
Pins 15	Base	100	2,911	100	0,066	100	0,020	100	
Pins 16	Base	100	2,836	100	0,092	100	0,020	100	
Pins 17	Base	100	2,895	100	0,099	100	0,019	100	
Pins 18	Base	100	2,736	100	0,068	100	0,021	100	
Pins 19	Base	100	2,669	100	0,094	100	0,019	100	
Pins 20	Base	100	2,718	100	0,095	100	0,019	100	
Pins 22	Base	100	1,562	100	0,041	100	0,014	100	Old 2
Pins 23	Base	100	1,569	100	0,042	100	0,015	100	
Pins 24	Base	100	1,567	100	0,040	100	0,015	100	
Pins 25	Base	100	1,549	100	0,040	100	0,015	100	
Pins 26	Base	100	1,652	100	0,037	100	0,015	100	
Pins 27	Base	100	1,676	100	0,034	100	0,018	100	
Pins 28	Base	100	1,654	100	0,041	100	0,016	100	
Pins 29	Base	100	1,649	100	0,036	100	0,015	100	
Pins 30	Base	100	1,719	100	0,043	100	0,017	100	
Pins 31	Base	100	1,699	100	0,033	100	0,015	100	
Pins 33	Base	100	1,628	100	0,041	100	0,020	100	Old 1
Pins 34	Base	100	1,558	100	0,042	100	0,016	100	
Pins 35	Base	100	1,623	100	0,042	100	0,017	100	
Pins 36	Base	100	1,601	100	0,037	100	0,017	100	
Pins 37	Base	100	1,595	100	0,043	100	0,014	100	
Pins 38	Base	100	1,567	100	0,040	100	0,013	100	
Pins 39	Base	100	1,634	100	0,034	100	0,017	100	
Pins 40	Base	100	1,614	100	0,044	100	0,015	100	
Pins 41	Base	100	1,665	100	0,036	100	0,016	100	
Pins 42	Base	100	1,646	100	0,042	100	0,014	100	

Layer thickness results of the single points. ~10 measurement point / pin



Sample pin	Ni [µm]		Pd [µm]		Au [µm]	
	Average	St dev	Average	St dev	Average	St dev
New 1	2,99	0,232	0,089	0,0125	0,020	0,0009
New 2	2,92	0,211	0,089	0,0139	0,020	0,0008
Old 2	1,63	0,062	0,039	0,0036	0,016	0,0010
Old 1	1,61	0,034	0,040	0,0033	0,016	0,0019
Limits	min 0,76 um		0,03-0,15 um		0,005-0,025 um	

Summary table of the XRF multipoint measurements

- In all three cases (Ni, Pd, Au), a significantly higher layer thickness can be measured on the new sample

EVC2600808 Measurement Notes

Sample comparison – summary

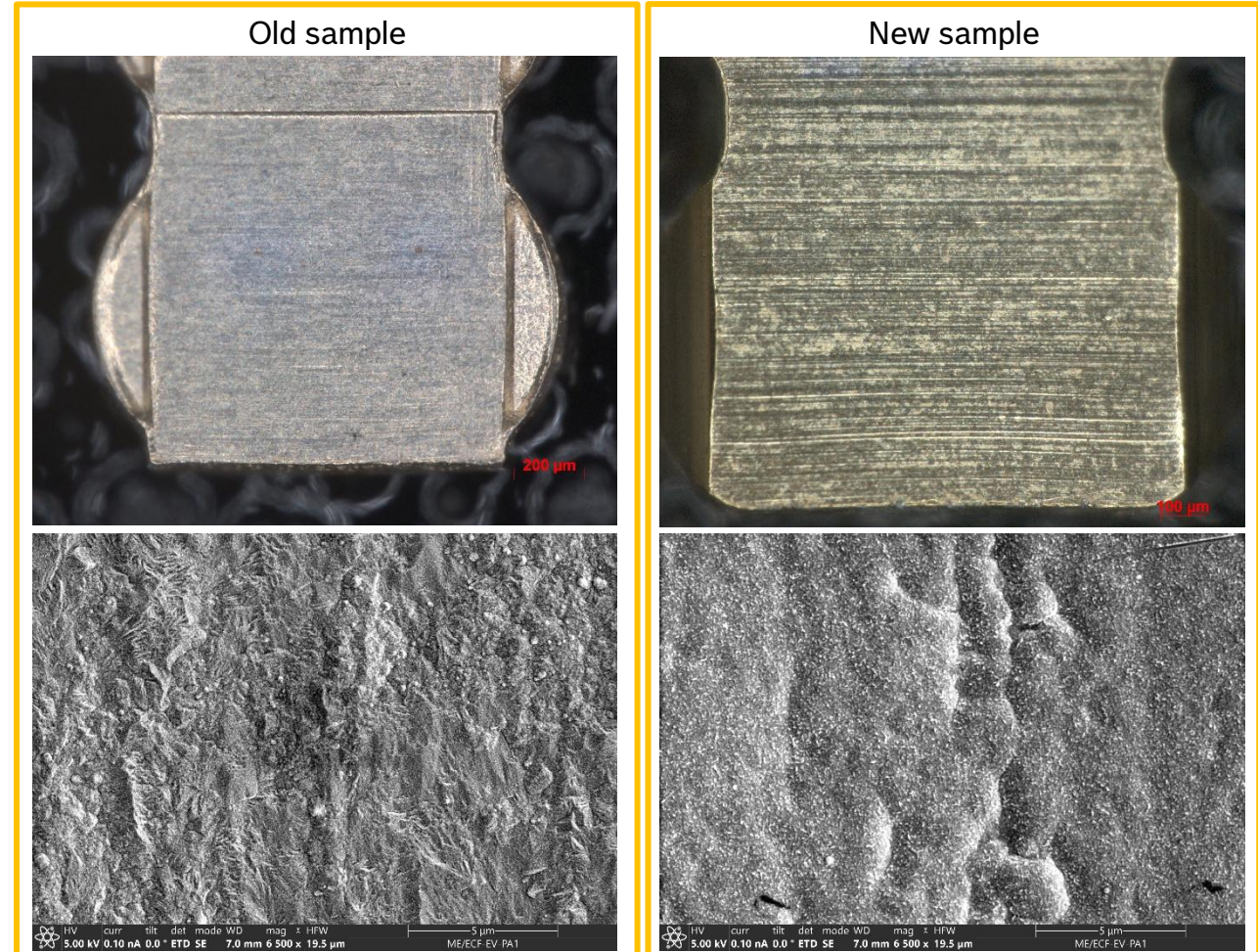
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	Old 1	1,61	0,034	0,040	0,0033	0,016	0,0019
FIB-SEM	New 1	3,119	0,215	0,0701	0,0033	0,0225	0,0014
	Old 1	1,773	0,081	0,0443	0,0021	0,0223	0,0012
Limits		min 0,76		0,03-0,15		0,005-0,025	

Layer thickness results based on FIB-SEM and XRF measurements

Sample pin	Ra (μm)	Rz (μm)
Old	0,54	2,86
New	0,67	4,08

Surface roughness values

- Most of the cases, a significantly higher layer thickness can be measured on the new sample
- The new sample shows higher values in surface roughness
- The surface of the new sample looks like more homogenous compared to old one



VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Affected Part Numbers / Plants & Customers

- Affected BOSCH PNs / customers / plants are provided via BoP Where-Used List (separate Excel attachment).

QueP	MrP	FleP	SgmP
7802376253	7802376253	7802376253	7802376253
7802376282	7802376279	7802376282	7802376279
7802376283	7802376282	7802376283	7802376282
7802376316	7802376283	7802376316	7802376283
7802376382	7802376302	7802376382	7802376302
	7802376303		7802376303
	7802376316		7802376316
	7802376335		7802376335
	7802376340		7802376340
	7802376348		7802376348
			7802376382

Customers
MB AG
Honda
AUDI
AMG
GROUP MB
MA
Ferrari
SL Italy
Maserati
VW
Honda R&D
Honda US
JLR
Ford
NISSAN
Porsche
Aston-M.
BMW
Bugatti
Volvo Car
BBAC (MB)
GM
BBA
BMW M GmbH
Toyota

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Difference Checklist

Affected field	Current Release State	Future State
Raw materials	Released material from defined and released suppliers	Changed. See material details in the IMDS comparison slide.
Bosch product IMDS	Released IMDS information	Changed. New IMDS entries will submit in the IMDS system.
Production equipment	Released machine No.: xxxxx	Not affected / No Change
Manufacturing process	Acc. to Control Plan	Not affected / No Change
Process parameters	Released process parameters	Not affected / No Change
Process controls	Released controls	Not affected / No Change
Quality Control	Released QC test plan	Not affected / No Change
Quality Control Equipment	Released QC Equipment	Not affected / No Change
Packaging equipment	Released machine No.: xxxxx	Not affected / No Change
Packaging parameters	Released process parameters	Not affected / No Change
Storage Conditions	Room Temperature / Normal	Not affected / No Change
Transport to Bosch	Released transport under controlled conditions	Not affected / No Change
Supp/y Chain	Released supply chain	Changed. New sub-supplier for the material

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Commercial Impact

	Bosch Other Divisions costs	Bosch VM costs	Bosch (VM) benefits
One time costs		3k EUR	Price down 0,03 EUR/pc Yearly estimated saving is cca. 50k Eur
Trial run costs		10k EUR (if needed)	
Validation costs			
PPAP costs		3k EUR	
Other costs			

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

MP-308.01 judgement - Customer involvement matrix

Technical impact of the change on		Reason / Details
Customer drawing	YES/ NO	NO
TCD	YES/ NO	NO
Fit / Form / Function	YES/ NO	NO
Design	YES/ NO	NO
IMDS declaration towards customer (Final execution only upon customer request)	YES/ NO	NO Minimal weight change only; therefore, there is no legal obligation to inform customers.
Change visible	YES/ NO	NO It is visible only in case of tear-down of the sensor unit.

Classification Line No.30c → Customer information required, if specified situation occurs at Bosch.

1) Does the change require an ECR process according to the MP-308.01-(CC) Realize Product and Process Changes in Series and Aftermarket Production?										Classification	
2) Does it affect special characteristics agreed with the customer?											
3) Does it affect a technical interface to the customer?											
4) Change type?											
↓	↓	↓	↓	5) Does it affect contract documents (e.g., specification sheet, customer drawing, offer drawing, TCD, data records, specifications, ...) or other serial documents provided to the customer (e.g., Control Plan, ...)							
				↓ 6) Does it affect fit, form, function, performance, reliability?							
Yes	No	No	change in the logistics	No	No	Testing	Expansion of the test without changing the method (e.g. increase in random sample) ^{1)ECR}	-	24		
Yes	No	No		No	No		Reduction/ cancellation of a not customer relevant testing (e.g. random sample) ^{1)ECR}	-	24a		
Yes	No	No		No	No	Relocation of production	Relocation of Tools within similar lines ^{1)ECR}	-	25		
Yes	No	No		No	No		Relocation of non-mobile equipment in a location without changing the process chain ^{2)ECR}	I	26		
Yes	No	No		No	No		Relocation of mobile equipment in a location without changing the process chain ^{2)ECR, 3)ECR}	-	26a		
Yes	No	No		No	No		Change in the location, relocation of the equipment, parallel production ^{4)ECR}	A	27		
Yes	No	No		No	No		Change in the location, relocation of the equipment, parallel production for roughing stages ^{3)ECR}	-	27a		
Yes	No	No	change in the logistics	Yes	→	New supplier, new sub-supplier	A	28			
Yes	No	No		No	Yes	New supplier, new sub-supplier	A	29			
Yes	No	No		No	No	New forwarding agent (carrier), external service provider, internal service provider ^{1)ECR}	-	30a			
Yes	No	No		No	No	New TIER2-Supplier for a purchased part	A	30b			
Yes	No	No		No	No	New ≥TIER3-Supplier for a purchased part ^{4)ECR}	I	30c			
Yes	No	No	change in the documents	No	No	Packaging to the customer, shipping, billing ^{1)ECR}	A	31			
Yes	No	No		No	No	Internal packaging (e.g. plant-plant, within plant,...) and delivered packaging ^{1)ECR}	-	31a			
Yes	No	No		Yes	→	Document adaptation to the condition of the released product	A	32			
Yes	No	No		No	No	Document adaptation to the condition of the released product or correction of formal error (ECR-correction)	-	33			
Yes	No	No		No	No	Change in non-product-related documents (e.g. from work instructions, accurate manufacturing instruction) ^{2)ECR}	-	34			
No	No	No					Re-use of tools after 12 or more months standstill	I	35		
No	No	No					Servicing/maintenance of existing tools or fast wearing tools	-	36		
No	No	No					Change in compiler version, Changing SW-tool without effect on customer-SW (Change in the debugger is not relevant here)	-	37		
No	No	No					Exchange with identical machine or with machine with identical process core, interchanging identical testing device	-	38		
A	Customer approval is required, if necessary, with initial sampling in customer agreement (no difference, if specified situation occurs at Bosch-supplier or Bosch)										
I	Customer information required, if specified situation occurs at Bosch										
-	Customer information not required										

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Risk Assessment – for changes without customer involvement

- Change visible in destructive tear-down: Yes, in case of tear-down of the sensor unit.
- Customer quality claims related to change: No
- Impact on design/process with known quality claims: No.
- Information exchanged with customers: potentially in later IMDS update.
- Overall risk: Low.

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Risk Mitigation

- **Traceability:**

- Via DMC

- **Measures to ensure delivery:**

- As it will be dual source, no risk from delivery point of view

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Risk Assessment

Risk category	Possible Risk (description)	Planned remedial measure	Remaining Risk Level ¹ (Low, Medium, High)
Quality/Spec.	Final product change	No change	Low
	Raw material plating change	Validation	Low
	IMDS change	Only quantity change	Low
	DBRFM	Risk discussed with Bosch	Low
Schedule	Run out of present stock	New tool build for selective bulding	Low
Supply	New supplier	Already validated and used supplier, dual source in the future	Low
Cost	Cost reduction	Cost reduction	Low

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

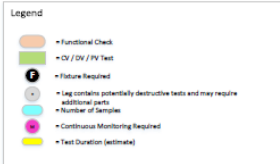
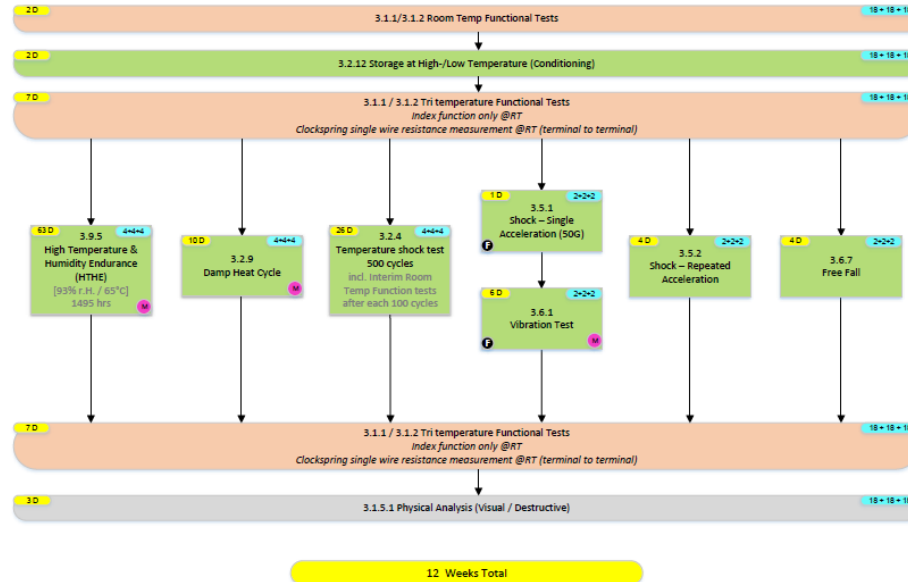
PV (Process Validation) and/or DV (Design Validation) validation proposal

BOURNS®
SENSORS DIVISION

DMS5 TIS/TOS & DMS5.2 TIS WBO plated terminal validation
HW-DMS_81_Testbeschreibung_7802-050-193_3V1 (extract 247+280+299 for Bourns)

Revision: 01
Revision Date: 23/09/2025

Total DUT distribution
In total 54 validation samples which consist of:
• 18 pos DMS5.2 MQ837W TIS produced in Ajka plant
• 18 pos DMS5 TIS Dual Channel produced in Tijuana plant
• 18 pos DMS5 TOS Dual Channel produced in Xiamen plant



Bourns Technical Program Mgr: Felix Römmelt Digital signature: Felix Römmelt, Date: 2025.09.23 14:19:32 +02'00'
Bourns Validation Engineer: Ulrich Zorell Digital signature: Ulrich Zorell, Date: 2025.09.23 14:19:32 +02'00'

Customer Project Engineer: Burak Altintas Digital signature: Burak Altintas, Date: 2025.10.20 09:01:10 +02'00'
Customer Product Manager: Ulrich Zorell Digital signature: Ulrich Zorell, Date: 2025.10.22 11:12:10 +02'00'

VM26_1034_Bourns DMS 5/5.2_Dual source for "Torque Terminal" and Reduce Pd/Au plating

Overall Schedule – INTERNAL

Schedule	From / To – Date	Comment / Status
Planning Kick-Off	CW 24	Attention: Involvement of all affected functions and plants
Team Review Approval	CW 26	Attention: All planning tasks closed
Product/Design Validation / Testing	ongoing	Attention: Supplier already started validations according the agreed plan.
Process Validation / Testing	CW 30	
Validation Report	CW 32	
Final Review Approval	CW 34	Attention: Consider appropriate lead time for approval
SOP Bosch Plant	CW 40	